Salivary acetaldehyde increase due to alcohol-containing mouthwash use: a risk factor for oral cancer.

Increasing evidence suggests that acetaldehyde, the first and genotoxic metabolite of ethanol, mediates the carcinogenicity of alcoholic beverages. Ethanol is also contained in a number of ready-to-use mouthwashes typically between 5 and 27% vol. An increased risk of oral cancer has been discussed for users of such mouthwashes; however, epidemiological evidence had remained inconclusive. This study is the first to investigate acetaldehyde levels in saliva after use of alcohol-containing mouthwashes. Ready-to-use mouthwashes and mouthrinses (n = 13) were rinsed in the mouth by healthy, nonsmoking volunteers (n = 4) as intended by the manufacturers (20 ml for 30 sec). Saliva was collected at 0.5, 2, 5 and 10 min after mouthwash use and analyzed using headspace gas chromatography. The acetaldehyde content in the saliva was 41 +/- 15 microM, range 9-85 microM (0.5 min), 52 +/- 14 microM, range 11-105 microM (2 min), 32 +/- 7 microM, range 9-67 microM (5 min) and 15 +/- 7 microM, range 0-37 microM (10 min). The contents were significantly above endogenous levels and corresponding to concentrations normally found after alcoholic beverage consumption. A twice-daily use of alcohol-containing mouthwashes leads to a systemic acetaldehyde exposure of 0.26 microg/kg bodyweight/day on average, which corresponds to a lifetime cancer risk of 3E-6. The margin of exposure was calculated to be 217,604, which would be seen as a low public health concern. However, the local acetaldehyde contents in the saliva are reaching concentrations associated with DNA adduct formation and sister chromatid exchange in vitro, so that concerns for local carcinogenic effects in the oral cavity remain.